Report
Of

A Full-Stack Development Internship at Thiranex:

Building Four Production-Grade Full-Stack Web Applications

(Portfolio Website, Task Manager, E-Commerce Application & Blog Platform)

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By
Roll Number of Student: 2410031017
Name of Student: SHIVEN VERMA
Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, Shiven Verma, do hereby declare that the internship report titled "A Full-Stack Development Internship at Thiranex: Building Four Production-Grade Full-Stack Web Applications" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Shiven

Signature
Student Name: Shiven Verma
Roll No.: 2410031017

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at Thiranex. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to Mr. Hariharan M, Founder & CEO of Thiranex, and the Thiranex team for the industry mentorship extended to me while I built and shipped four full-stack web applications over the course of this internship.

I also thank the School of Computer Science and Engineering, IILM University, and my faculty coordinator for arranging and supervising this internship, and for their continued guidance throughout the program.

I express my gratitude to my parents for their blessings.

Shiven

Signature
Student Name: Shiven Verma
Roll No.: 2410031017

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Chapter 3: Internship Completion Certificate

Thiranex
SKILL DEVELOPMENT & FUTURE TECH

Date: 30 July 2026

TO,

Shiven Verma

Intern ID: THX-JUL3026-475

Subject: Offer Letter for Internship in Full Stack Development

Dear **Shiven Verma,**

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	**Intern - Full Stack Development**
Commencement Date	**30 Jul 2026**
Completion Date	**29 Aug 2026**
Work Mode	**Remote / Project-Based**

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

Thiranex

This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in

Offer Letter, Thiranex — Intern ID: THX-JUL3026-475 (Full Stack Development)

Thiranex
SKILL DEVELOPMENT & FUTURE TECH

Certificate

OF ACHIEVEMENT

This acknowledgment is proudly presented to

Shiven Verma

This is to certify that **Shiven Verma** has successfully completed an internship in **Full Stack Development** from **30 Jul 2026** to **29 Aug 2026**.

VERIFIED CERTIFICATE
ID: THX-JUL3026-475

Hariharan M
Founder & CEO

Certificate of Achievement, Thiranex — Full Stack Development (30 Jul 2026 – 29 Aug 2026)

Chapter 4: Project Description

4.1 Introduction

This report covers my internship in Full Stack Development at Thiranex, undertaken remotely on a project basis from 30 July 2026 to 29 August 2026. Rather than a single product, the internship was structured around four independent, production-grade web applications, each covering a different domain — a personal portfolio site, a task management application, an e-commerce storefront, and a blogging platform with comments — so that I would get hands-on practice with the full web stack (frontend, backend, database, and authentication) repeated across different data models and user flows instead of just once.

Across the four builds I worked with React (Vite) and Tailwind CSS on the frontend, Node.js and Express on the backend, and Prisma ORM over PostgreSQL, MySQL, or SQLite depending on the project, along with JWT-based authentication and REST API design. This report explains the organisation I interned with, the problem each project addresses, the architecture and technologies used, the way I approached building all four, and the outcomes for each.

4.2 Organization Profile

Thiranex is a skill development and future-tech focused organisation that runs remote, project-based internships for students, pairing them with industry mentorship across domains including full stack development. Internships are structured around a defined role, a fixed commencement and completion date, and periodic performance review, with successful completion recognised through a formal certificate of achievement.

I interned with Thiranex as an Intern – Full Stack Development (Intern ID: THX-JUL3026-475), working remotely on a project basis for one month, from 30 July 2026 to 29 August 2026.

4.3 Problem Statement

Classroom coursework in a CSE program typically teaches frontend, backend, and database concepts in isolation — a React component here, an SQL query there — without requiring a student to own a complete, deployable application end to end: designing the schema, securing the API, connecting it to a real UI, and handling the edge cases that only show up once a project is used rather than graded. Employers and real products, in turn, expect exactly that ownership across the stack.

The internship set out to close that gap by requiring four separate full-stack applications, each with authentication, persistent storage, and a complete set of CRUD flows, so that the underlying patterns (auth, data modelling, API design, state management, protected routes) would be practiced repeatedly across different problem domains rather than learned once and forgotten.

4.4 Project Objectives

- Build a full-stack personal portfolio website with a dynamic, database-backed project showcase instead of static content.
- Build a task management web application with authenticated, per-user task CRUD, dashboard analytics, and multi-parameter filtering.
- Build an e-commerce web application covering product catalogue, cart, checkout, order tracking, and role-based (admin/user) access.
- Build a blogging platform with full post CRUD, an interactive commenting system, and a creator dashboard.
- Apply consistent, production-style practices across all four — JWT authentication, password hashing, input validation, and a documented REST API — rather than tutorial-level shortcuts.
- Gain practical experience with Prisma ORM across three different databases (PostgreSQL, MySQL, and SQLite) and with deployment-ready project structure.

4.5 Scope of the Project

The internship's scope covered four self-contained deliverables, each due and submitted on a separate milestone date and marked completed:

- Personal Portfolio Website (due 06 Aug 2026) — full-stack portfolio with a dynamic project showcase.
- Task Management Application (due 13 Aug 2026) — CRUD task manager with authentication and dashboard analytics.
- E-Commerce Web Application (due 20 Aug 2026) — online store with product management and order tracking.
- Blog Platform with Comments (due 27 Aug 2026) — blogging platform with posts and a comment system.

Native mobile apps and third-party payment gateway integration (beyond a basic checkout flow) were kept out of scope; the emphasis throughout was a correctly modelled backend, a secured REST API, and a responsive React frontend for each application.

4.6 Technologies and Tools Used

The technology choices were kept consistent where possible across projects, so that patterns learned on one carried over to the next, while each project's database and a few libraries were chosen to fit its specific needs. The stack is summarised in Table 4.1.

Table 4.1: Technologies and Tools Used

Layer	Technology	Purpose / Notes
Frontend Framework	React 18/19 (Vite)	Component-driven UI across all four applications
Styling	Tailwind CSS	Utility-first, responsive design system
Backend Framework	Node.js + Express	REST API layer and routing for every project
ORM	Prisma ORM	Type-safe schema and queries over the relational database
Database	PostgreSQL / MySQL / SQLite	PostgreSQL (Blog, E-Commerce), MySQL (Portfolio), SQLite (Task Manager)
Authentication	JWT + bcrypt/bcryptjs	Token-based sessions and salted password hashing
Validation	Zod / express-validator	Request-level input validation on the API
State & Routing	React Router, Context API	Auth/cart state and protected client-side routes
Notifications & Icons	React Hot Toast, Lucide React	User feedback and consistent iconography
Deployment Targets	Vercel, Render	Frontend and backend hosting used across the projects

4.7 System Architecture

All four applications follow the same underlying pattern: a React single-page frontend communicates over a REST API with an Express backend, which in turn talks to a relational database through Prisma ORM (or a direct MySQL connection pool for the portfolio site). JWT tokens issued at login are attached to subsequent requests and verified by backend middleware to protect routes and enforce per-user data isolation. Figure 4.1 illustrates this shared architecture.

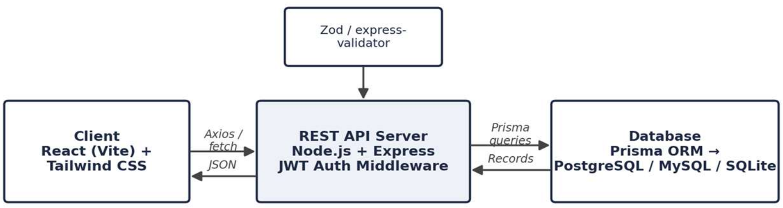

Figure 4.1: Common client-server-database architecture shared across the four applications

On top of this shared foundation, each project layers its own domain logic: the Task Manager adds status/priority-based filtering and dashboard aggregation computed from live records; the E-Commerce application adds a cart and order pipeline sitting on User, Product, Cart, CartItem, Order, and OrderItem models with a separate admin area; the Blog Platform adds ownership-based authorization so only an author can edit or delete their own posts, plus comment moderation for post owners; and the Portfolio site exposes a simple public projects API that the frontend falls back to static data for if the backend is unavailable.

4.8 Methodology

For each of the four projects, I followed the same build loop: design the data model in Prisma first, build and test the REST API endpoints, then build the corresponding React screens against that API, and finally add the cross-cutting concerns — authentication, validation, and error handling — before treating the feature as complete. Where relevant, an automated test suite was added directly against the API (for example, the Blog Platform includes an integration test suite covering registration, auth, post CRUD and ownership checks, search/filtering, and dashboard aggregation; the Task Manager's backend ships 17 passing integration tests).

All four projects were tracked as separate milestones with fixed due dates over the internship period (06, 13, 20, and 27 August 2026) and were each submitted on track and marked completed.

4.9 Expected Outcomes

Figure 4.2: The four full-stack applications built during the internship

Full-Stack Portfolio Website

A full-stack personal portfolio built with React (Vite) and Tailwind CSS on the frontend and Node.js/Express with a MySQL connection pool on the backend. Instead of hardcoding projects into the page, the site fetches them from a /api/projects endpoint backed by a seeded MySQL database, with a graceful fallback to static data if the API is unavailable. The outcome was hands-on experience integrating a frontend, backend, and database for a live, deployable project, and a working seed script that provisions the database and its six showcased projects on setup.

Task Management Application (TaskFlow)

A production-style task manager built with React (Vite), Tailwind CSS, Express, and Prisma. It implements registration and login with bcrypt password hashing (12 salt rounds) and JWT-based authorization, strict per-user data isolation, full task CRUD with title, description, priority, status, and due date, an inline status-update control with optimistic UI updates, debounced search, multi-parameter filtering (status, priority, due date) and sorting, dynamic overdue detection, and a dashboard with live statistics computed from the database. The backend ships with a 17-test integration suite (Jest/Supertest) covering auth and task logic. This delivered practical experience in full-stack application structure, protected-route design, and dynamic, database-driven UI state.

E-Commerce Web Application

An online store built with React, Tailwind CSS, Express, and Prisma over PostgreSQL, modelling Users, Products, Carts, Cart Items, Orders, and Order Items. It covers product browsing and detail pages, a cart and checkout flow, order history and order-detail views, user authentication and profile management, and a separate admin area for product/order management, with JWT authentication and express-validator input validation on the API. This gave hands-on experience building a more complex, multi-entity full-stack application with real-world e-commerce features, including cart and order state management on the frontend.

Blog Platform with Comments (InkFlow)

A full-stack blogging platform ("InkFlow") built with React 19, Vite, Tailwind CSS v4, Express 5, and Prisma over a Supabase-hosted PostgreSQL database. It supports registration and login with JWT and bcrypt, full post CRUD with a live side-by-side preview while writing, categories and tags, draft mode, an interactive commenting system with ownership-based edit/delete and comment moderation for post owners, full-text search with category/tag filtering and sorting, a creator dashboard with post/comment metrics, and rate limiting on auth endpoints against brute-force attempts. An automated integration test suite verifies registration, login, protected routes, post/comment CRUD and ownership, search and filtering, and dashboard aggregation. This project gave the deepest experience with content-management style features, ownership-based authorization, and user interaction at scale.

All four applications were submitted on their respective due dates within the internship window and marked completed successfully. Beyond the individual features, the internship as a whole gave me repeated, hands-on practice designing relational schemas with Prisma across three different databases, securing REST APIs with JWT, and building responsive React frontends that consume those APIs — the core full-stack skill set the internship was designed to build.

4.10 Certificates and Communication Proof

The Internship Offer Letter and the Certificate of Achievement issued by Thiranex, signed by Mr. Hariharan M, Founder & CEO (Intern ID: THX-JUL3026-475), are attached earlier in this report (Chapter 3).

The source repositories for all four applications are retained and available as supporting proof:

- Full-Stack Portfolio Website — github.com/shivenverma/Full-Stack-Portfolio-Website
- Task Management Web Application (TaskFlow) — github.com/shivenverma/Task-Management-WebApp
- E-Commerce Web Application — github.com/shivenverma/E-Commerce-Application
- Blog Platform with Comments (InkFlow) — github.com/shivenverma/Blog-Platform

Chapter 5: Bibliography / References

[1] Shiven Verma, “Full-Stack-Portfolio-Website (source repository).” [Online]. Available: https://github.com/shivenverma/Full-Stack-Portfolio-Website

[2] Shiven Verma, “Task-Management-WebApp (source repository).” [Online]. Available: https://github.com/shivenverma/Task-Management-WebApp

[3] Shiven Verma, “E-Commerce-Application (source repository).” [Online]. Available: https://github.com/shivenverma/E-Commerce-Application

[4] Shiven Verma, “Blog-Platform (source repository).” [Online]. Available: https://github.com/shivenverma/Blog-Platform

[5] Thiranex, “Skill Development & Future Tech.” [Online]. Available: https://www.thiranex.in

[6] Vercel Inc., “Deployment Documentation.” [Online]. Available: https://vercel.com/docs

[7] Meta Open Source, “React Documentation,” React. [Online]. Available: https://react.dev

[8] Prisma, “Prisma ORM Documentation.” [Online]. Available: https://www.prisma.io/docs

[9] Tailwind Labs, “Tailwind CSS Documentation.” [Online]. Available: https://tailwindcss.com/docs

[10] OpenJS Foundation, “Express.js Documentation.” [Online]. Available: https://expressjs.com